THE CONTRACTION OF A RATIONAL BRIDGE PULLS BACK THE DUALIZING SHEAF ISOMORPHICALLY

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file curves.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: the category equivalence with its quasi-inverse, the restriction sequence on the bridge with the author's small-detail remark on the degree count, the vanishing input, the commuting square of functors, and the Yoneda conclusion are all retained verbatim.

Editorial note. The label lemma-rational-bridge-canonical names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Allowed standard prerequisites (editorial).

- Dualizing module by representability. Same chapter, Lemma lemma-duality-dim-1-CM; Duality for Schemes, Lemma duality-lemma-dualizing-module-proper-over-A. Characterizes omega_X as the coherent module representing F -> Hom_k(H^1(X, F), k), the representability framework in which the conclusion is drawn.
- Picard behaviour under the contraction. More on Morphisms, Lemma more-morphisms-lemma-bijection-on-Pic; Cohomology, Lemmas cohomology-lemma-projection-formula and cohomology-lemma-apply-Leray. Gives essential surjectivity of c^* on the relevant subcategories, the projection formula c_*c^*N = N, R^1c_*c^*N = 0, and the Leray comparison of the two H^1-dual functors.

Source reference. Original lemma 001P: sources/stacks-chow/tags.txt, source lines 70–70. Prerequisite pointer invoked at proof line(s) 6429; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 01F4: sources/stacks-chow/tags.txt, source lines 1857–1857. Prerequisite pointer invoked at proof line(s) 6416; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 01E8: sources/stacks-chow/tags.txt, source lines 1825–1825. Prerequisite pointer invoked at proof line(s) 6397; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 0AWP: sources/stacks-chow/tags.txt, source lines 13727–13727. Prerequisite pointer invoked at proof line(s) 6382; it names a standard

Date: stacks-project:curves.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 0E3M: sources/stacks-curves/pinned-original-source.tex, source lines 6201–6201. Prerequisite pointer invoked at proof line(s) 6369; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 0E34: sources/stacks-curves/pinned-original-source.tex, source lines 1012–1012. Prerequisite pointer invoked at proof line(s) 6407; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 0BS3: sources/stacks-curves/pinned-original-source.tex, source lines 864–864. Prerequisite pointer invoked at proof line(s) 6380; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 0C6M: sources/stacks-curves/pinned-original-source.tex, source lines 2005–2005. Prerequisite pointer invoked at proof line(s) 6409; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 0E24: sources/stacks-chow/tags.txt, source lines 17594–17594. Prerequisite pointer invoked at proof line(s) 6395; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Lemma 0.1. *Let $c : X \to Y$ be the contraction of a rational bridge (Example 0E3M). Then $c^*\omega_Y \cong \omega_X$.*

Proof. You can prove this by direct computation, but we prefer to use the characterization of ω_X as the coherent $\mathcal{O}_X$-module which represents the functor $Coh(\mathcal{O}_X) \to Sets$, $\mathcal{F} \mapsto \mathrm{Hom}_k(H^1(X, \mathcal{F}), k) = H^1(X, \mathcal{F})^\vee$, see Lemma 0BS3 or Duality for Schemes, Lemma 0AWP.

To be precise, denote $\mathcal{C}_Y$ the category whose objects are invertible $\mathcal{O}_Y$-modules and whose maps are $\mathcal{O}_Y$-module homomorphisms. Denote $\mathcal{C}_X$ the category whose objects are invertible $\mathcal{O}_X$-modules $\mathcal{L}$ with $\mathcal{L}|_C \cong \mathcal{O}_C$ and whose maps are $\mathcal{O}_Y$-module homomorphisms. We claim that the functor

$$c^* : \mathcal{C}_Y \to \mathcal{C}_X$$

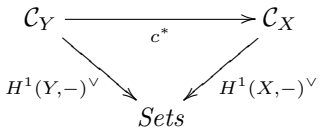
is an equivalence of categories. Namely, by More on Morphisms, Lemma 0E24 it is essentially surjective. Then the projection formula (Cohomology, Lemma 01E8) shows $c_*c^*\mathcal{N} = \mathcal{N}$ and hence c^* is an equivalence with quasi-inverse given by c_*.

We claim ω_X is an object of $\mathcal{C}_X$. Namely, we have a short exact sequence

$$0 \to \omega_C \to \omega_X|_C \to \mathcal{O}_{C \cap X'} \to 0$$

See Lemma 0E34. Taking degrees we find $\deg(\omega_X|_C) = 0$ (small detail omitted). Thus $\omega_X|_C$ is trivial by Lemma 0C6M and ω_X is an object of $\mathcal{C}_X$.

Since $R^1c_*\mathcal{O}_X = 0$ the projection formula shows that $R^1c_*c^*\mathcal{N} = 0$ for $\mathcal{N} \in \mathrm{Ob}(\mathcal{C}_Y)$. Therefore the Leray spectral sequence (Cohomology, Lemma 01F4) the diagram



of categories and functors is commutative. Since $\omega_Y \in \mathrm{Ob}(\mathcal{C}_Y)$ represents the south-east arrow and $\omega_X \in \mathrm{Ob}(\mathcal{C}_X)$ represents the south-east arrow we conclude by the Yoneda lemma (Categories, Lemma 001P). □